

Annals of Operations Research, 108, 111–122, 2001
© 2002 Kluwer Academic Publishers. Manufactured in The Netherlands.

An Optimization Based Approach to the Train Operator Scheduling Problem at Singapore MRT

KIM-LIN CHEW and JOHNNY PANG
Indepth Consulting

QIZHANG LIU, JIHONG OU and CHUNG-PIAW TEO
Department of Decision Sciences, Faculty of Business Administration, National University of Singapore

Abstract. Singapore Mass Rapid Transit (SMRT) operates two train lines with 83 kilometers of track and 48 stations. A total of 77 trains are in operation during peak hours and 41 during off-peak hours. In this article we report on an optimization based approach to develop a computerized train-operator scheduling system that has been implemented at SMRT. The approach involves a bipartite matching algorithm for the generation of night duties and a tabu search algorithm for the generation of day duties. The system automates the train-operator scheduling process at SMRT and produces favorable schedules in comparison with the manual process. It is also able to handle the multiple objectives inherent in the crew scheduling system. While trying to minimize the system wide crew-related costs, the system is also able to address concern with respect to the number of split duties.

Keywords: crew scheduling, optimization, bipartite matching, tabu search, set partitioning model

1. Introduction

In 1972, the Singapore Government and the United Nations Development program, with the International Bank for Reconstruction and Development, embarked on a major transport study in Singapore. The study considered traffic demand forecasts, impact of ongoing and committed road improvements, impact of proposed administrative pricing, traffic engineering and other environmental restraints unique in Singapore contexts. After examining five alternative public transport systems, it reached the conclusion that a conventional rail mass transit system was a superior alternative to complement the existing public bus service. This forms the major impetus for several more phases of feasibility studies to identify potential locations and design of the new mass rapid transit system. The Singapore MRT Limited (SMRT) was finally incorporated on August 1987 to operate the MRT system, under a license granted by the Land Transport Authority of Singapore.

SMRT currently operates two train lines with 83 kilometers of track and 48 stations. The two train lines are: the East-West Line goes from Pasir Ris (the easternmost station) to Boon Lay (the westernmost station), and the North-South Line goes from Marina Bay in the south up north to Woodlands then down to Jurong East in the west.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

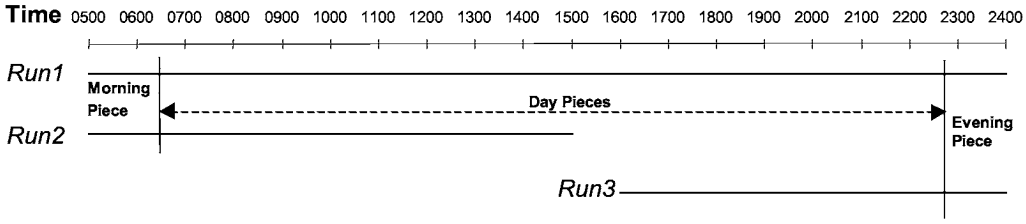


Figure 1. Typical train runs.

A total of 77 trains are in operation during peak hours and 41 during off-peak hours. Due to different traffic demand patterns, there are three types of train schedules, one for weekdays, one for Saturdays and one for Sundays and public holidays. The train schedules consist of train runs as shown in figure 1. Some of the train runs are almost a day long, like run 1, while others are rather short, like runs 2 and 3. The three types of schedules change from time to time as the traffic demand changes. The train operator scheduling problem is to generate a duty schedule for train operators to man all the train runs in a given train schedule. The duty schedule is constrained by many labor rules as in a typical crew scheduling problem. The objective is to minimize the total crew-related cost (given by a complicated calculation that will be explained in detail later) of manning the given train schedule.

Previously an experienced manager was in charge of scheduling train operator duties. He accomplished his job manually by tapping onto his past experiences to make minor changes to existing crew schedules. It takes the manager plus an assistant more than one week to come up with a feasible operator schedule every time the train schedules are changed (even slightly). With the imminent changes in the transport scene in Singapore (e.g., recent government initiatives include building a new train line, proposed merger between train and bus companies, etc.), SMRT management decided that a computerized system to generate train operator schedules was a necessity and called upon the first two authors to study the problem and to design a customized software package for their special two-line MRT system. A team of researchers from Indepth Consulting and National University of Singapore were subsequently engaged in the development of the optimization engine for the scheduling problem.

Attempts to develop computer programs to compile bus and crew schedules began in the late 60s. Many examples of early systems are reported in [10]. Most of the early systems are based on some heuristic modeling and solution approaches, which are surveyed in Wren [11]. Such approaches typically follow a two-staged procedure. In the first stage, the bus (train, flight) lines are cut into small pieces of work; and the second stage involves how to optimally combine these pieces to generate feasible duties covering all the lines with the minimal cost. Bodin et al. [2] provides an excellent survey of the research in this field, and examples of computer systems following optimization-based approaches are reported in Desrochers and Rousseau [7].

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

In the 90s, advancement in computational tools makes solving very large-scale crew scheduling problem a reality. See Beasley and Cao [1] for a recent literature survey on this topic. However, the basic two-stage procedure has been the norm in problem modeling and solution approaches. The optimization problem formulation is then solved by various algorithmic techniques, including implicit enumeration procedures of branch-and-bound types, Lagrangian relaxation approaches, column generation approaches, and others. Beasley and Cao [1] gave examples of these solution approaches. We also refer the readers to the comprehensive web resources collected in http://www.informatik.hu-berlin.de/~weinert/CrewSched.html for many other references in the crew scheduling literature, which we have to regrettably omit due to space constraint.

For the SMRT problem, we adopt the same 2-stage procedure: we first subdivide the train runs of a given train schedule into smaller pieces of work (as in the manual process), and then try to optimally combine these pieces of work into legal train operator duties that cover all the train runs. The second step is formulated as a mathematical optimization problem. There are two such formulations, one for the overnight duties and one for the day duties. The night-duty model is basically a bipartite matching model. It pairs late evening pieces of work with early morning pieces of work (for next day). The day duty problem is much more difficult. It is formulated as a set-partitioning problem, and near optimal solution is found by a tabu search algorithm. Tabu search has been known as an "extremely promising" heuristic algorithm for solving large-scale combinatorial optimization problems in a number of practical settings [5]. It was first suggested by Glover [8] and since then many exciting implementations of tabu search have been reported for various application problems, like production planning and scheduling, time-tabling, resource allocation, vehicle routing and scheduling, etc. See the references in Glover and Laguna [9].

The optimization engine, coupled with a user-friendly window GUI software package, is now currently in use at SMRT. The package greatly simplifies the train operator scheduling process for the company, reducing the time to generate a feasible operator schedule from a week (under the manual system) to several minutes, and the schedules generated are almost always favorable over those generated manually. The package is also very flexible in that it allows human intervention at any point of the automated scheduling process.

In the following we describe our approach to the train operator-scheduling problem at SMRT. In section 2, we describe in details the nature of our problem, the cost structure deemed important according to SMRT management, and the constraints stipulated by the local labor rules. Section 3 presents the mathematical formulations and the solution algorithms. We then describe in detail the software package in section 4 and the experiences and lessons learned during the implementation stage.

2. Problem specifications

The first stage of work-piece cutting cuts the train runs into work pieces. One of the labor rules requires that day duties cannot start before 0645 hours and end after 2315 hours.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

The train runs before 0645 hours and after 2315 hours have to be manned by overnight duties. We therefore first cut out the beginning piece on a train run that starts early in the morning from its start to the first possible shift-changing point after 0645 hours, as shown on runs 1 and 2 in figure 1. Similarly, we cut out the ending piece on a train run that ends late in the evening from the last shift-changing point before 2315 hours to the end, as shown on runs 1 and 3 in figure 1. These beginning and ending pieces will be paired to form night duties manned by train operators who stay overnight in the company. The remaining train runs are cut into consecutive pieces of work that are not more than 270 minutes each. 270 minutes is the maximum length of time an operator is allowed to man a train continuously without a break. These work pieces are used to form day duties.

In addition to these normal work pieces, there are a few train runs where an operator is needed at each end of the train as it turns around at terminal stations. This is to enable fast turning around of the train at the terminals during peak hours, and is called *double-ending* duties at SMRT. Double-ending pieces are formed by identifying all double-ending cases, and then by grouping them into pieces that can be handled by double-ending operators. There are complex rules governing the forming of double-ending pieces. For instance, under normal circumstances double-ending can be two-way, i.e., from the terminus to the adjoining station or vice versa. However if the train is withdrawing to a depot or is just entering service, double-ending has to be one-way.

As far as crew scheduling is concerned, double-ending pieces are handled in the same way as other work pieces. There are various restrictions, however, on the formation of a duty:

Split duties. The two types of day duties, *normal* and *split*, are differentiated by the length of the meal break and the spread of the work pieces. The constraints on these two types of duties are shown in table 1.

Table 1
Time constraints on duties.

	Day duty		Night duty
	Normal	Split	
Working hours:			
Maximum	9 hours	8.5 hours	9 hours
Minimum (unofficial)	6 hours	6 hours	6 hours
Spread:			
Maximum	10 hours	12 hours	10 hours
Minimum (unofficial)	6.75 hours	–	6.75 hours
Part duty:			
Maximum	4.5 hours	4.5 hours	4.5 hours
Minimum	Nil	Nil	Nil
Break between workpieces:			
Minimum	45 mins	>1 hours	45 mins
Maximum	60 mins	5 hours	60 mins

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

Other restrictions include

(a) a meal-break time window – meal breaks must be taken not earlier than 2 hours and not later than 5 hours after reporting for duty;

(b) a user-specified limit on the number of pieces per duty; and

(c) a user-specified limit on the number of duties beginning at a particular crew point.

In addition, the user wants to have the option of specifying whether split duties are allowed in the duty schedule that is generated.

Traveling time allowances. If a train operator is on a *normal* duty and resumes duty at another location after a break, the traveling time between the previous location and the location where he resumes duty will be counted as part of his working hours. However if he is on a *split* duty and resumes duty at an *equivalent* location, the time taken to travel to the equivalent location will not be counted as part of his working hours. There are also time allowances in minutes for various activities relating to reporting for work and finishing work at the depots and the stations.

The objective. SMRT is interested primarily in minimizing the total number of duties, but would also like to minimize the number of split duties as a secondary objective. We propose a duty costing formula that is used as an objective function in our optimization formulations to accurately reflect the concerns of SMRT management. The formula adds up the following cost components.

- Duty cost: Associated with each duty is a constant cost.
- Split duty cost: Each split duty will be charged an extra split duty cost so as to minimize number of split duties.
- Balance cost: Each duty is charged a balance cost proportional to the square of the number of working pieces in the duty.
- Working time cost: Desired working time per duty is 7.5 hours, and thus, a duty with total working time less than 7.5 hours incurs a cost proportional to the time short.
- Idle time cost: The idle time during the breaks of a duty is charged a cost proportional to the total break time.
- Idle traveling cost: When an operator travels on a train as a passenger while on duty, a cost is levied with a fixed rate.
- Shift allowances: There are allowances for duties between 1500 hours and 2300 hours and the night shifts that cover the late evening and early morning pieces.

Among the above cost components, the duty cost overwhelms the others so that the first criterion of the model is to minimize the number of duties formed. The balance cost is defined in such a way that the tabu search algorithm finds duties with fewer pieces. Note that the cost parameters are fictitious penalties created for the search algorithm and do not reflect the actual cost associated with the duties.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

3. Problem formulations and solution techniques

As stated above, the night duties and the day duties are contained in two separate schedules. Below we present our problem formulations to generate optimal schedules and the solution techniques applied to the formulations.

3.1. Night duty scheduling

A night duty always consists of a morning piece and an evening piece. Thus, the night-duty scheduling problem is formulated as a weighted bipartite matching problem, with all the morning pieces in one group and the evening pieces in the other. The solution to the problem is obtained using the well-known Hungarian algorithm. The choice is made based on the algorithm's simplicity in structure and ease in programming.

3.2. Day duty scheduling

A day duty can be composed of many separate work pieces. Initially, SMRT Management conveyed to the authors the intention of using only 2-piece duties for ease of control and management. Thus, we proposed a weighted non-bipartite matching formulation and implemented the *shortest augmenting path* method developed by Derigs [6] for its solution. After seeing the solution to such a formulation, SMRT management wanted the system to include 3-piece duties to see the impact on improvement to the solution. The matching problem formulation no longer applies, and we had to consider a set partitioning problem or a set covering problem formulation. Both formulations are much more difficult to solve than the non-bipartite matching problem. Among the two, it is generally true that a set covering problem formulation is easier to solve. However, a solution from a set covering problem formulation needs to be manually adjusted to eliminate some over-covering. When it was tried out for the train operator scheduling problem at SMRT, it was found that the over-covering is so much that the number of duties increases substantially after manual modification. Allowing double covering in the duties was also ruled out. Although this will result in fewer duties being generated (as the model is relaxed from set partitioning to set covering), it will be hard to persuade the train operators to accept these unnecessary duties, and padding unnecessary work in between work pieces separated by long breaks in the duties might be perceived as an indirect way to circumvent the labor rules.

Therefore we adopted a set partitioning problem formulation as given below:

$$
\begin{aligned}
\text{(SPP)} \quad \text{Minimize} \quad & \sum_{d \in D} c_d y_d \\
\text{subject to} \quad & \sum_{d:\, i \in d} y_d = 1 \quad \text{for } i \in P, \\
& y_d \in \{0, 1\} \quad \text{for } d \in D.
\end{aligned}
$$

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

P is the set of work pieces and D is the set of all feasible duties. c_d is duty d's cost, and $y_d = 1$ if duty d is in the solution and 0 otherwise. A work piece $i \in d$ if it is covered by duty d.

We tested several techniques to solve SPP, including branch-and-bound approaches and Lagrangian relaxation approaches. These techniques were found to be far too slow for the size of the input of the actual problem. They were not able to solve the problem to optimality, and the best solution generated using these approaches were often not as good as those generated manually. We also realize that the actual problem is multi-criterion in nature. The desires of SMRT management included minimizing the number of duties and also the number of split duties. We finally took up a tabu search based approach for its ease of implementation, and also for the fact that the multi-criterion nature of the problem can be used to good effect to guide the direction of search in the execution. With this approach we were able to solve successfully SPP for up to 500 work pieces and the number of feasible duties possibly running into millions.

In essence, tabu search is an algorithm of local search augmented with adaptive memory. Starting with an initial solution, the algorithm proceeds iteratively by replacing current solution s with a *best* neighbor $s' \in N(s)$ among all possible moves. This procedure stops when certain stopping criteria are satisfied. The neighborhood $N(s)$ of the current solution s is defined by move operators, which play the key role in tabu search algorithms. The move operators determine the operations that can be done to move from one solution to another. In our tabu search algorithm, to solve the set partitioning problem as formulated above, there are three move operators: *cross operator*, *exchange operator*, and *relocate operator*. The cross operator swaps work pieces of two duties by first cutting the duties into 2 segments each, and recombine them to form two new duties. The exchange operator takes one work piece out from two duties and exchanges their roles to form two new duties. The relocate operator relocates one work piece from one duty to another duty. See the illustration of figure 2.

Tabu search algorithms differ from other descent search methods in the introduction of a *tabu list*. In our implementation, the tabu list is a two dimensional array $\theta_x = (\tau_{ij})$ in which τ_{ij} stores tabu active status of edge ij. Edge ij is *in solution* if work

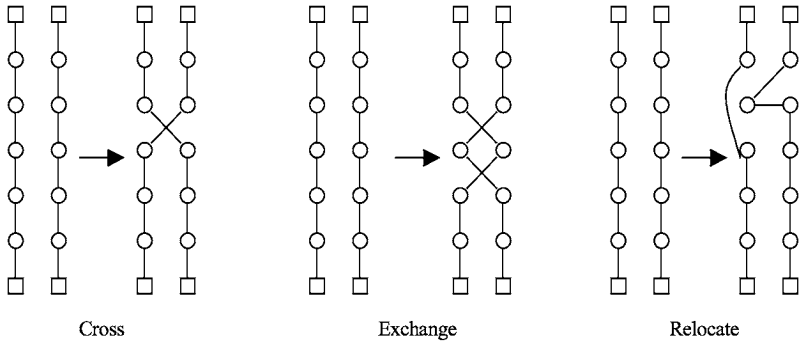


Figure 2. Move operators.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

piece i and work piece j are adjacent in a duty of the solution, and *out of solution* if otherwise. Specifically, τ_{ij} is the time (i.e., iteration number) when tabu-active status of edge ij ends, i.e.,

$$\tau_{ij} = start_{ij} + tenure,$$

where $start_{ij}$ is the iteration number immediately before edge ij entered or left a solution and *tenure* is the tabu tenure, which is a fraction of the size of θ_x. In practice, tabu search is effective when *tenure* is a random number in the range $[|P|^2/10, |P|^2/9]$ generated at iteration $start_{ij}$. Moves that change status (in or out of solution) of tabu active edges are prohibited during the *tenure*.

Remarks.

(1) The size of the neighborhood of a solution is $O(|P|^2)$. Typically in tabu search, instead of searching the whole neighborhood, most implementations follow one or the other kind of *candidate list* strategy to evaluate only those moves with bigger potentials. For our problem, however, due to time allowances, labor rules and the complexity of the cost structure, checking feasibility and evaluating a candidate list itself is quite expensive. Therefore, after extensive experiments, we decided to maintain a dynamic duty pool that stores a small portion of duties in current solution, and apply move operators to duties in the pool in each iteration. When no improvement can be made, new duties from current solution will be randomly selected to replace some duties in the pool.

(2) The quality of the solutions obtained depends on several measures. To facilitate a meaningful comparison, we formulate a total cost function of the solution by aggregating the various measurements together (i.e., using the various cost components as listed in section 2). In the actual implementation of the tabu search algorithm we experimented with different choices of the proportionality parameters to aggregate the various measurements while computing the total cost of the solutions. Since the ultimate objective is not to minimize any costs but rather the number of duties and also the number of split duties, we change the proportionality parameters to these costs intelligently when the tabu search algorithm gets stuck at some region of the solution space. A more systematic approach is to generate the solution via hierarchical multi-objective method, rather than just changing the weights of cost parameters. However, our computational results show that the proposed approach is already able to generate very good solution for the SMRT problem instances.

Caprara, Fischetti and Toth [3] developed a very efficient heuristic method (CFT algorithm) for solving crew scheduling in the Italian Railway Company. They formulated the problem as a set covering problem, and tackled it based on Lagrangian relaxation with subgradient optimization. As the SCP model provides a reasonable lower bound to the set partitioning model adopted in this paper, we compare our solutions with those obtained by CFT algorithm to show the quality of our tabu search algorithm.

In CFT algorithm, firstly, a pool of all feasible duties is generated subject to the constraints stated above. Then a dynamic pricing scheme, similar to column generation

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

Table 2
Performance comparison.

Case		CFT				Tabu search	
		#Duties (SCP)	#Duties (SPP)	Size	CPU time	#Duties	CPU time
North-South Line	Saturday	73	81	264 × 418463	611	77	122
	Sunday	68	77	237 × 432671	587	70	94
	Weekday	89	98	297 × 848642	1835	94	231
East-West Line	Saturday	64	72	219 × 237643	879	70	67
	Sunday	53	63	236 × 297734	918	59	62
	Weekday	106	118	314 × 1073665	1624	110	181

used for solving large-scale LP, is adopted such that in all phases the subgradient optimization and a greedy algorithm run only on a small subset of columns. For more details of the algorithm, see Caprara, Fischetti and Toth [3]. For the test cases provided by SMRT, there are about 220–300 rows. We do the comparison under the assumption that no split duties are allowed in both models, mainly because the machine at SMRT disposal are not able to handle the millions of columns when split duties are allowed. Even so, the SCP model still generates about 200,000 to 1,000,000 columns. Our comparison focuses only on the number of duties. We run both algorithms on a PC with a Pentium 600 MHz processor. The result is shown in table 2, where the CPU time is recorded in seconds. The fourth column lists the numbers of duties after the over-coverings in the SCP solutions are removed by a greedy enumeration algorithm.

The solutions generated by our tabu search algorithm are close to those generated by the SCP model of CFT, but are better than the SPP solutions directly obtained from the solutions of CFT. Furthermore, the speed and ability to handle problems with split duties make tabu search an ideal approach to tackle this problem.

4. Implementation

In this section we present the computer implementation of our approach to the train operator scheduling problem at SMRT and report on the implementation and testing experiences. The computer implementation results in a software package (MrtSchedule) which is structured as follows:

- Scheduling Manager is a bridge connecting GUI with optimization engine. It links GUI with data manager, scheduling engine and the result;
- Data Manager interprets data files to basic system objects: Train Runs, Relief Points and Working Pieces, which are input to scheduling engine;
- Scheduling Engine implements the tabu search algorithm to obtain schedule;
- Solution includes a list of duties, which are represented as in figure 3.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

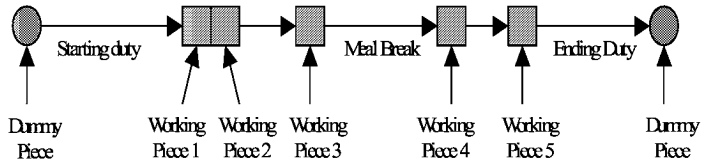


Figure 3. Duty representation.

Table 3
Comparison of the results obtained for the North-South Line.

Schedule type	Manual system	Automated system	Difference
Weekday	83 (with 17 split)	80 (with 15 split)	3 (4%)
Saturday	83 (with 8 split)	72 (with no split)	11 (13%)
Sunday	82 (with no split)	61 (with no split)	21 (26%)

Table 4
Comparison of the results obtained for the East-West Line.

Schedule type	Manual system	Automated system	Difference
Weekday	88 (with 17 split)	84 (with 19 split)	4 (5%)
Saturday	74 (with 13 split)	64 (with 3 split)	10 (14%)
Sunday	71 (with no split)	56 (with no split)	15 (21%)

A GUI was added to allow users to view the optimization process and to decide whether the result is good enough to terminate the program. The software was tested at SMRT with actual train schedules, and the results were very positive. Benefits were observed in the following aspects:

- *Time required to prepare duty schedules from scratch.* This has been reduced from 22 weeks to three weeks. In fact, the time required to generate a duty schedule is usually less than 6 hours for weekend schedules and less than a day for weekday schedules. The rest of the time is spent on fine-tuning the schedules to cater to train operators' preferences.
- *Quality of the schedules.* The number of duties in computer generated schedules on average is 10% to 20% less than that in manually prepared schedules for Saturday and Sunday train schedules and 5% less for weekday train schedules. The actual figures are listed in tables 3 and 4. In addition, the number of split duties has been significantly reduced in most cases, which means that the quality of the solution has improved significantly.
- *Reporting capability.* The computer software can display and print train schedules and duty schedules graphically. This was not possible in the past when the staff had to manually cut and paste the train runs to form the operator schedules.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

- *Training staff to perform crew scheduling.* The computer package makes it easy to train new staff to take over the crew scheduling function. A new staff just needs to learn how to use the software to do some fine-tuning.
- *Time saving from double-ending pieces.* The Crew Schedule program automatically assigns double ending pieces, resulting in a saving of one week of work.
- *Data portability.* The transfer of data from the train timetable schedule compiler to the crew schedule program is now carried out electronically instead of manually typing in. This process, which takes about a week in the past, are now done in a matter of minutes.

The computer system as implemented is thus clearly superior to the manual process. But before the more efficient duty schedules can be implemented, several issues pertaining to the operation need to be addressed. One of them is that the present crew rostering system adopted by the SMRT results in equal number of men on duty every day of the week. However, under the new schedule, fewer staff is needed on weekends than on weekdays as can be clearly seen from tables 3 and 4. Therefore before it could implement the more efficient duty schedules, SMRT would need to address the impact of the new schedule on its rostering system. Another concern is whether the train operators will accept the schedule produced by the system, and whether there are any other genuine concerns of the operators that have not been taken care of in the system. SMRT has thus opted to reduce gradually the weekend manning levels, by manually adjusting the output from the scheduling system. This is also done in consultation with the train operators to uncover further hidden constraints in the train operations. This approach has helped to win over the support of the train operators, who now felt that they "owned" the system and are not "controlled" by the system.

5. Conclusion

In this paper, we report on an optimization-based approach to develop a computerized train-operator scheduling system that has been successfully implemented at SMRT. Combinations of optimization techniques ranging from a traditional matching algorithm to a new tabu search heuristic were employed in the approach. The approach we have adopted for the problem is novel in the sense that the multi-criterion nature of the problem is being used to good effect in the local search approach. This also allows us to circumvent the difficulty of solving the set partitioning formulation using traditional mathematical programming approach, which does not seem to be able to address the issues of multiple criteria effectively, since solving the large-scale model with a single objective function is already difficult and slow. Although the theoretical efficiency of our proposed approach cannot be established analytically, the drastic improvement over the current manual system suffices to demonstrate once again the applicability of the powerful OR approach in tackling real business issues. In fact, one of the most satisfying experiences from this study is to influence and help SMRT management to see

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

the power and importance of this approach in tackling their operational problems. Furthermore, the train operators are in no way being displaced by the computer – the extra operators needed under the old manual system were gradually re-assigned to the more effective computer schedule as SMRT increases the number of train runs. In fact, SMRT is currently operating with close to 100 train operators.

References

[1] J.E. Beasley and B. Cao, Dynamic programming based algorithm for crew scheduling, Computers and Operations Research 25 (1998) 567–582.
[2] L. Bodin, B. Golden, A. Assad and M. Ball, Routing and scheduling of vehicles and crews; the state of the art, Computers and Operations Research 10 (1983) 63–211.
[3] A. Caprara, M. Fischetti and P. Toth, A heuristic method for the set covering problem, Operations Research 47 (1999) 730–743.
[4] L. Cavique, C. Rego and I. Themido, Subgraph ejection chains and tabu search for the crew scheduling problem (1997) submitted.
[5] Committee on the Next Decade of Operations Research (CONDOR 1998), Operations research: The next decade, Operations Research 36.
[6] U. Derigs, Solving non-bipartite matching problems vis. shortest path techniques, Annals of Operations Research 13 (1988) 225–261.
[7] M. Desrochers and J.-M. Rousseau, *Computer-Aided Transit Scheduling* (Springer, Berlin, 1990).
[8] F. Glover, Heuristics for integer programming using surrogate constraints, Decision Sciences 8 (1977) 156–166.
[9] F. Glover and M. Laguna, *Tabu Search* (Kluwer Academic, 1997).
[10] J.-M. Rousseau, *Computer Scheduling of Public Transport 2* (North-Holland, Amsterdam, 1985).
[11] A. Wren, *Computer Scheduling of Public Transport: Urban Passenger Vehicle and Crew Scheduling* (North-Holland, Amsterdam, 1981).

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.